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Declaration:

I have read and understand Appendix 2 in the Student Handbook: "Some advice on the avoidance of plagiarism".

I hereby declare that the attached report is exclusively my own work, that no part of the work has previously been submitted for assessment (although do note that material in "Interim Report" may be re-used in the final "Project Report" as it is considered part of the same assessment), and that I have not knowingly allowed it to be copied by another person.

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Abstract:

The aim of this project is to investigate how the statistics of the cosmic microwave background (CMB) depend on cosmological parameters and how future observations of the CMB will change our understanding of cosmology, hence, the project title 'extracting cosmology from cosmic microwave background data'. The first part of this project will focus on how the power spectra changes for varying cosmological parameters. A good theoretical base will be established then existing code will be modified to carry out this task. The second half of this project will focus on the analysis of a hypothetical futuristic data set which will demonstrate what improvements to cosmology could be made through the development of new observatories. A Markov chain Monte Carlo algorithm will be developed and implemented to achieve this.

Introduction:

The beginnings of the cosmic microwave background:

The CMB was discovered in 1964 by Robert Wilson and Arno Penzias. However, it is important to understand the ideas of the universe at the time. Two main theories of cosmology existed, the steady state model [1] and the hot big bang. In the modern day, the big bang model is widely accepted. The discovery and further analysis of the CMB was one of the main contributors to the acceptance of the big bang model.

Before the CMB was discovered, those in favour of the big bang model predicted that there should be photons left over from the beginning of the universe and were working on finding the temperature they should be at present day. Gamow [2] and collaborators predicted they should be able to measure a black body spectrum with temperature between 5K and 50K. However, they had not yet observed these relic photons.

Robert Wilson and Arno Penzias had built a radio telescope to observe radio transmission from satellites. This was the most advanced receiver for the time; however, they thought something was wrong with their instrumentation. No matter the direction, the telescope always detected noise with a much higher intensity than expected. They investigated every possible cause of this noise, even cleaning pigeon droppings from the horn, but the noise persisted. They measured it to be at 3.5K.

Penzias then, with the guidance of B.F. Burke a radio astronomer, alerted Robert Dicke of Princeton university, whose colleague Jim Peebles had written a paper predicting cosmic background radiation [3]. After confirming his discovery and performing follow up measurements, R.H. Dike, P.J.E. Peebles, P.G. Roll and D.T. Wilkinson published a paper reporting the discovery of the CMB [4] while Penzias and Wilson publish a paper describing their measurement of excess antenna temperature. [5]

Our current understanding:

We now understand the CMB to be the photons released from the first time the universe became transparent (or recombination). Before this time, the universe was so hot and dense that atoms could not form. In this plasma, photons would most commonly interact with electrons through Compton scattering (more specifically the low energy limit of Compton scattering – Thompson scattering). Electrons, when encountered by a photon, absorb that photon and re emit at the same wavelength but, in a different direction. The photon is scattered. Because the electron density was high, photons would not travel far before scattering again. In other words, the Compton mean free path was small, smaller than the particle horizon. The universe was opaque for this reason and cannot be observed. This scattering meant that photons were coupled to electrons which were in turn coupled with protons

through coulomb attraction. The consequence of this was the photons of the early universe were coupled with matter and so it behaved as a photon-baryon fluid.

Once the temperatures dropped low enough, to around 3000K, atoms formed allowing photons to travel freely without scattering. This is known as recombination and is the first time we can observe the universe. These first photons released are the photons we observe when measuring the CMB and inform us on the distribution of matter in the early universe because of their previous coupling to it. We now measure these photons to be at 2.7K, the microwave regime, though they were much hotter to begin with. The expansion of the universe stretched the wavelength of these relic photons, decreasing their temperature.

When measuring the temperature of the photons that make up the CMB we see slight fluctuations across the sky. These fluctuations are caused by the slight changes in the distribution of matter in the early universe and we see the same fluctuations in the temperature because photons are coupled with matter.

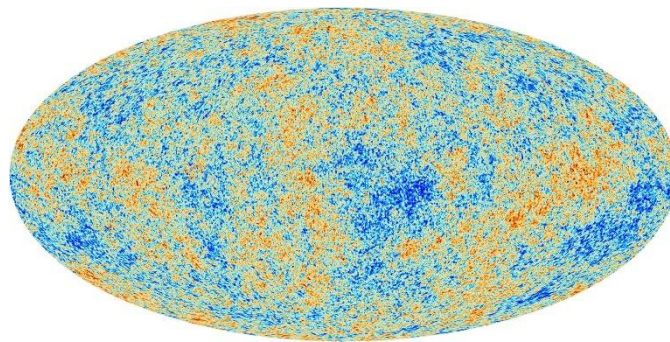


Figure 1: temperature map of the CMB measured by the European space agency's Planck satellite. Blue spots show temperatures colder than the average and red spots show temperatures hotter than average. Structure can be seen at both the large scale and small scale.

From figure 1 it can be seen how areas of the sky deviate in temperature from the mean of 2.7K. Though Planck, the most recent satellite to have measured the CMB, was able to produce an extremely detailed image, these deviations are, in fact, very small, on the order of micro-Kelvin. The areas of higher density in the early universe go on to form structure and eventually the stars and galaxies present today.

How we analyse the CMB:

To gain a better understanding of the structure in the CMB the difference in temperature between two points separated by an angle on the sky is measured. This is done over the whole sky and an average is taken. We then vary this angle to measure the small scale and large-scale structures. this information is represented by plotting the amplitude of the temperature difference against the angle on the sky or more specifically against the multipole moment which is proportional to one over the angle on the sky, see figure 2.

When looking at the correlation in temperature between two points in the sky we call this the temperature-temperature, or TT , power spectrum. This is not the only way we measure correlations in the CMB. We also consider how the polarisation of the CMB varies spatially; we will discuss this later.

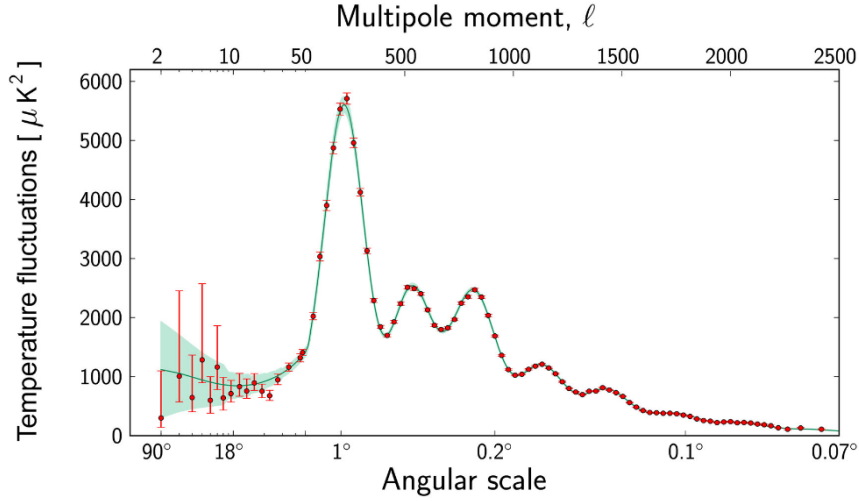


Figure 2: the TT power spectrum of the CMB as measured by the European space agency Planck mission [6].

We use the power spectra to determine cosmological parameters which allow us to infer key quantities of the universe such as its age, shape and whether it is expanding or not. From this information, we can build models of the universe.

The structure of this graph is caused by acoustic oscillations in the early universe. Before recombination, matter (and by extension photons) would fall in and out of potential wells. The gravitational attraction between matter would pull it into dense regions. The matter pulled photons with it and, as the density increases, the radiation pressure caused by these photons increases counteracting the gravitational in fall. This balance between gravitational force and radiation pressure creates a harmonic motion of matter and photons falling in and out of potential wells. This happens on both the largest and smallest scales. We see these waves as hot and cold spots on the sky when measuring the CMB and on the power spectra, we see them as peaks.

The peaks of figure 2 can tell us about the conditions of the early universe so it is important to understand the physics behind the graph. Mathematically, the relationship between photons and baryons can be described as,

$$\frac{\Delta T}{T} = [\Theta + \phi + \hat{r} \cdot \vec{v}]$$

Where Θ represents intrinsic density fluctuations, ϕ represents the gravitational component and $\hat{r} \cdot \vec{v}$ represents the Dopler term. $\frac{\Delta T}{T}$ represents the correlation in temperature between two points in the sky, $\hat{r}$ is the comoving distance to the surface of last scattering and v is the fluid velocity.

To understand the evolution of the intrinsic fluctuations both the gravitational term and the baryon evolution equation must first be solved.

$$\ddot{\delta} + 2H\dot{\delta} + (c_s^2 k^2 - 4\pi G \bar{\rho})\delta = 0$$

This equation describes the behaviour of baryonic matter in the early universe and is derived from the continuity equation of mass conservation, Euler's equation for a fluid, Poisson equation for mass and gravity, entropy conservation and the equation of state for density and pressure. Where c_s is the speed of sound, G is the gravitational constant, $\bar{\rho}$ is the mean density and δ is given by:

$$\delta = \frac{\rho(\vec{x}, t) - \bar{\rho}(t)}{\bar{\rho}(t)}$$

The baryon evolution equation has the solution:

$$\Theta + \phi = \frac{1}{3}(1 + 3R) \cos(kc_s\eta) - R\phi$$

This is nothing more than the equation for an oscillator with a driving force. The Doppler term is given by the continuity of energy equation,

$$\hat{r} \cdot \vec{v} = \frac{(1 + 3R)}{3\sqrt{3(1 + R)}} \phi \sin(kc_s\eta)$$

Which is out of phase with the intrinsic fluctuations and gravitational term. The combination of these two equations plus a silk dampening effect produces the shape of the power spectra seen in figure 2.

Measurements of the CMB:

We do not measure the power spectra of the CMB with one instrument but instead use multiple telescopes, both ground based and satellite. Ground based observatories have large mirrors which we can't get to space so are able to measure the smallest angular separation. However, they are limited to only half the sky so suffer at larger angular scales to gain enough measurements. Satellites have access to the whole sky but have smaller mirrors. This makes them suited to larger angular separation which have less points available on the sky.

COBE (cosmic background explorer) was the first satellite to measure the CMB and extract the power spectra. The differential microwave radiometer on board was only able to measure the largest of angular scales with the ability to measure multipoles of $l \leq 30$ with the error bars of measurements for $l \geq 10$ being dominated only by cosmic variance [7]. Cosmic variance refers to how, at large scales in the universe, there are a limited number of points available to be sampled. When measuring the correlation between two points separated by large angles, there are fewer points available than when measuring this for small angular separation. This makes the measurements at small multipoles more statistically uncertain and so we cannot determine them more certainty than this. Theoretically, this could be overcome by measuring the CMB from another point of the universe, for example, a different galaxy. The measured CMB would be slightly different while still occurring from the same event, the big bang, so would increase the sample size reducing the cosmic variance. However, it is not possible to travel to another galaxy and take this measurement so the small multipole region of the power spectrum will remain limited by cosmic variance.

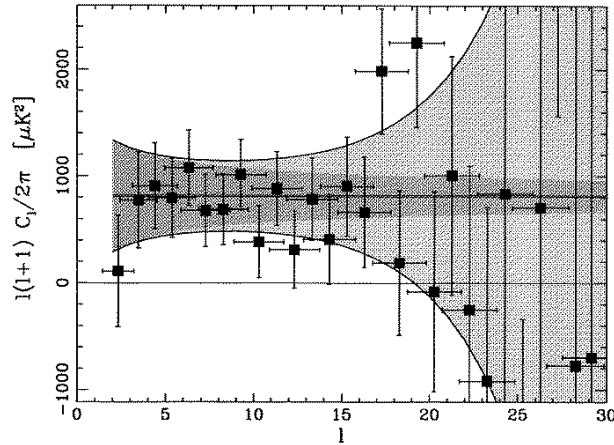


Figure 3: obtained from the 4-year NASA COBE data, the graph shows the power spectra of the CMB at small multipoles or large angular separation. [7]

In figure 3 It can be seen how COBE was only able to probe small multipoles. In this region, there is not much structure as the distances are larger than the horizon scale. Information hasn't had enough time to be exchanged so regions are not causally connected. The conditions here are caused by the very early universe and are not affected by acoustic oscillations. This region is known as the Sachs-Wolfe plateau.

Though the data from COBE is perhaps less impressive than future missions, it was paramount in proving that there were fluctuations in the temperature of the CMB and that those fluctuations show structure. This discovery was so significant that it was awarded the 2006 Nobel prize in physics.

The next generation satellite was WMAP (Wilkinson microwave anisotropy probe) which was able to measure much smaller angular separation.

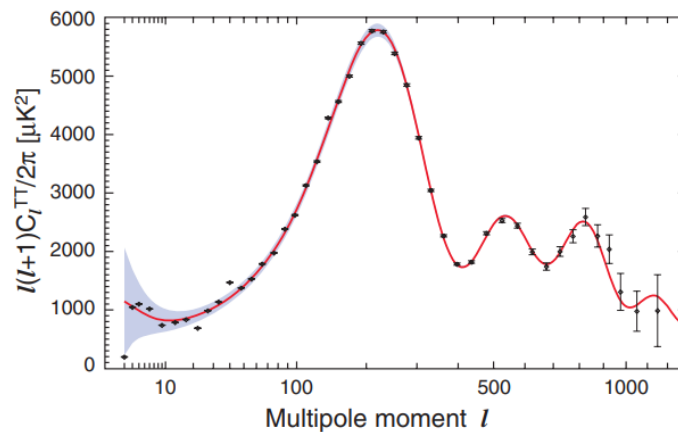


Figure 4: data from the NASA WMAP probe used to plot the power spectra of the CMB reaching much smaller angles than COBE. [8]

Figure 4 highlights how WMAP was able to reveal the structure in the temperature-temperature power spectrum. Multipoles larger than $l \sim 100$ show the acoustic oscillations in the proton baryon fluid.

Planck was the most recent satellite to measure the power spectra. Figure 2 shows how Planck managed to reach smaller angles or larger multipoles. To obtain the very best set of data we combine satellite data with data obtained from ground-based experiments.

Polarisation:

Up until this point, only the variations in temperature have been discussed. However, this is not the only way fluctuations in the CMB can be seen. Because of the specific motion of the proton-baryon fluid, quadrupolar temperature distributions are created. This means that when photons are scattered off electrons, the electrons radiate in a preferred direction. The cooler direction. Perfect thermal radiation is not polarised. We only see this polarisation because of the motion of the plasma in the early universe.

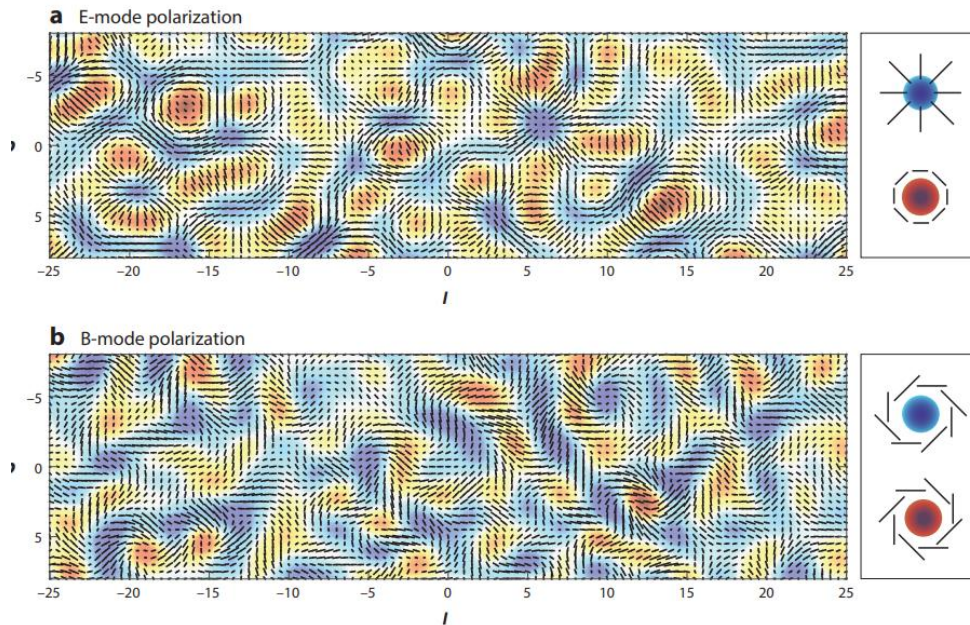


Figure 5: representation of the polarisation pattern for E-modes, top, and B modes, bottom. The top right shows how E-modes behave around a hot, red, and cold, blue, spot and the bottom right shows the same for B-modes. Credit: Marc Kamionkowski and Ely D. Kovetz [9]

We see polarisation in two modes, E and B. Figure 5 shows how E and B modes are similar to electric fields and magnetic fields respectively, hence their names. Density perturbations only cause E modes where gravitational waves cause both E and B modes.

When referring to the power spectra of the CMB, this does not only include the temperature-temperature, TT , spectrum but also the temperature-E mode polarisation, TE , and E mode-E mode polarisation, EE , spectra. Similarly to how the TT spectrum compares the temperature of two points, the EE spectrum compares the E-mode polarisation between two point and the TE spectrum compares the temperature and E-mode polarisation of two points.

DASI, degree angular scale Interferometer, was the first to detect polarisation fluctuations in the cosmic microwave background in 2002 [10]. WMAP then improved on this detection by measuring the temperature-polarisation power spectra of the CMB seen in figure 6.

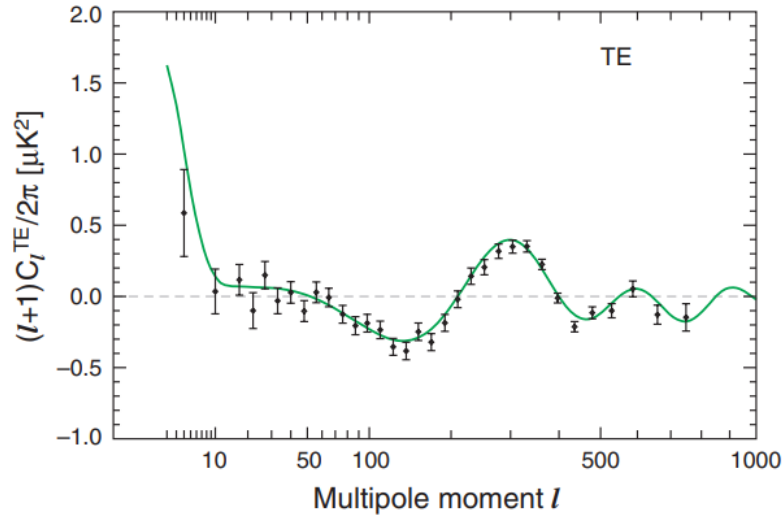


Figure 6: temperature-polarisation (TE) power spectrum measured by NASA's WMAP [8]. The green line shows the best fit to the Λ CDM model, discussed in next section.

The Planck satellite gives us the most up to date measurement of the polarisation power spectra.

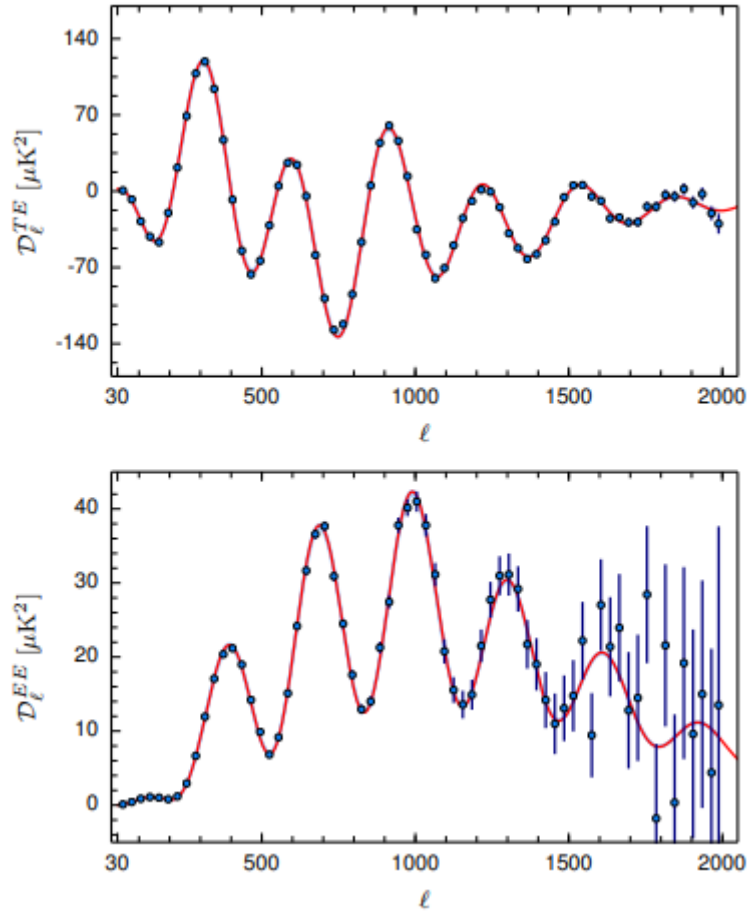


Figure 7: TE power spectrum, top, and EE power spectrum bottom as measured by the European space agency's Planck satellite [11].

The improvement made by the Planck satellite can be seen by comparing figure 6 and 7. As we will discuss in the next section, the temperature-temperature power spectrum is useful to determine cosmological parameters. Because the polarisation power spectra measure the same parameters as the TT spectrum, they can be used to break degeneracies between parameters.

Though the TT power spectrum is extremely well constrained, we have not achieved this with the polarisation power spectra. The next steps in this field involve fully characterising E-mode polarisation and detecting the B mode polarisation from gravitational waves. This is something scientists have yet to develop the instrumentation to measure.

Interpreting the power spectra:

The power spectra of the CMB holds information about the structure of the universe and by understanding what causes each of the peaks we can then constrain important cosmological parameters which determine our understanding of the universe. Recall how temperature fluctuations are caused by matter falling in and out of potential wells. Wells occurring at dense regions and hills at empty regions. These potential wells and the matter oscillating in and out of them occur at all scales. From the largest to the smallest. On the largest scale we see matter that has just had enough time to compress once before recombination. We see this in the first peak of the power spectra and the sound horizon can be inferred from it. The sound horizon is the furthest distance sound could have travelled during the age of the universe at that time and is a useful tool in cosmology for measuring distances.

The next peak shows matter that's just had time to fall in then back out again and the peak after that shows matter that's just had time to fall in out then back in again. These modes of oscillation are harmonic with each other. As we will see the position and size of these peaks are related to cosmological parameters.

When creating a theory of cosmology, the model must predict the peaks of the power spectra. The best model we have is Λ CDM or the standard model of cosmology. Lambda represents the cosmological constant and is included in the name as it suggests the presence of dark energy. CDM stands for cold dark matter. The Λ CDM model includes dark energy, dark matter and baryonic matter in our universe. As the name suggests, the standard model of cosmology fits both the data from the CMB and other sources very well. Any competing model must predict the power spectra better than Λ CDM. This is a tall order, though cosmologists are trying to find a better model or tweak Λ CDM so that they can better describe the universe. Λ CDM is not perfect. Dark energy and dark matter, though a hot topic for debate, are not understood fully so finding a better fit than this model could reveal insights in this area.

Λ CDM describes an expanding universe beginning with a big bang and has the presence of cold dark matter. Small quantum fluctuations in the early universe cause both the CMB fluctuations and in the modern-day distribution of galaxies according to this model [12]. Six parameters are used to describe the universe in Λ CDM. These are Ω_m or Ω_c , Ω_b , h , n_s , τ and σ_8 or A_s . Ω_c and Ω_b are the density parameters for dark matter and baryonic matter respectively, h is the Hubble constant, n_s is the scalar spectral index, τ is the optical depth to reionisation and A_s is the amplitude of fluctuations. The density parameters Ω_c and Ω_b describe the ratios of dark matter and baryonic matter to the total matter content of the universe. The Hubble constant describes how fast galaxies move away from each other. By tracing the motion of these galaxies back through time, we can infer that they all originated from a single common point, the big bang. The Hubble constant can therefore give a measure of how old the universe is. Scalar spectral index, n_s , describes how density fluctuations vary with angular scale. Reionisation is the period of the universe after the recombination when stars and galaxies reionise the neutral hydrogen atoms. A larger value of τ gives galaxies a longer time to evolve. Changing the values of any of these parameters will change the predicted shape of the power spectra.

Λ CDM predicts the power spectra of the CMB very well, in fact, in figure 2 and 7, the points lie very closely to the best fit line, especially for the TT spectra in figure 2 and the TE graph in figure 7. Once it gets hard to see how well data fits a model, we consider a different kind of plot known as a residual plot. A residual is the difference between an observed data point and theory.

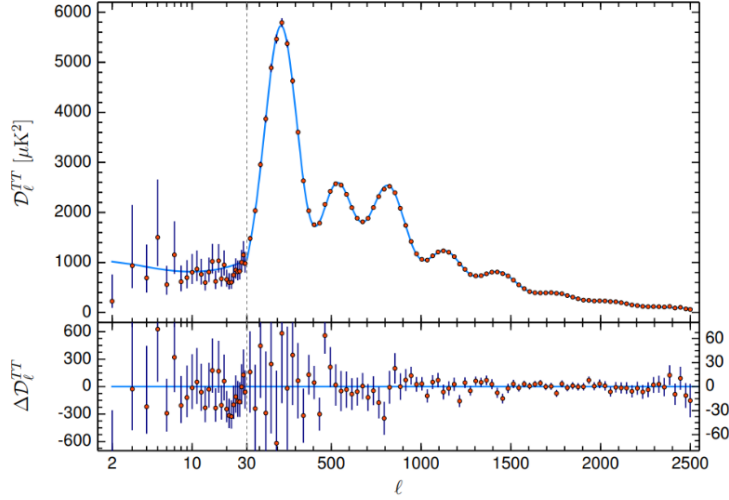


Figure 8: TT power spectra in top plot with data points from Planck and a best fit line based on Λ CDM. The bottom plot shows the residuals. Taken from Planck 2018 results [13].

A residual plot makes it clear how far from the model data points lie. The residual plot in figure 8 highlights how the points of the TT power spectra appear to fit very closely to the model. There is more scatter at higher multipoles though this is caused by cosmic variance as discussed previously. Residual plots make it clear how well a data set fits a model and so is a useful tool when comparing models. There is a small dip in the residuals around multipole 30 which could be interpreted as a deviation from the Λ CDM model. However, because it is only a small discrepancy and resides in the low multipole, cosmic variance limited area of the power spectrum, cosmologists don't consider this a contradiction to Λ CDM.

Problems with Λ CDM

Λ CDM has been extremely successful. It has stood up too many tests, Planck being a recent high precision example. The power spectra from Planck agree very well with this model. The data seen in figures 2 and 7 are fitted to Λ CDM with each spectra producing consistent values of each parameter. The model also does a good job of predicting the formation of large-scale structure and cosmic abundances of energy and matter. There are some tensions, however, with other data such as type Ia super nova measurements.

This is not the only problem which has arisen with Λ CDM. There is no explanation as to what dark matter or the cosmological constant are. The cosmological constant is particularly problematic as it has given rise to one of the biggest problems in physics, the vacuum catastrophe, a 120 orders of magnitude disagreement between observation and theory [14].

Because of its success in predicting the CMB power spectra, cosmologists are hesitant to start from scratch when finding ways to tackle the problems with Λ CDM. Instead, extensions to the model are often proposed. These extensions often focus on the parameters held constant in Λ CDM. These include the sum of neutrino masses, the physical neutrino density parameter, the equation of state of dark energy and the total density parameter among others. The consideration of the first two parameters has led to a

group of hot dark matter (HDM) theories being proposed. These propose the neutrino as the primary HDM particle and for a time there was debate as to whether the universe was dominated by hot or cold dark matter. It is generally accepted now that HDM does not dominate as this type of universe is not compatible with observations of galaxy clusters [15]. More modern theories including HDM will contain a combination of hot and cold dark matter which is often referred to as mixed dark matter.

Any proposed theory of cosmology must predict the power spectra of the CMB, and it must do it as well, if not better than Λ CDM does to be considered a competitor. Often theories that don't derive from Λ CDM struggle with this which is why Λ CDM has been so successful for so long.

Using the CMB to test cosmology.

Because of the importance of the CMB power spectra to cosmological models, to gain a better understanding of cosmology and perhaps find a better model of cosmology, we must measure the power spectra as precisely as possible. Planck was the most recent telescope to make a leap in precision measurements of the CMB improving on WMAP's smallest measurement of around multipole 1000 to a multipole of 2500 revealing the damping tail of the TT power spectrum. Though this area of the TT power spectrum is less dramatic than the peaks of smaller multipoles, it corresponds to scales smaller than the mean free path of photons before recombination.

Recombination was not an instantaneous process and because of this, as temperatures cooled close to the ionisation temperature of hydrogen the photon mean free path increased as the coupling between baryons and photons became less perfect. This larger mean free path smooths out the fluctuations in CMB temperatures at this scale. This results in the exponential damping tail. The tail is sensitive to Ω_c and Ω_b . The first few peaks of the TT power spectrum are also sensitive to these parameters, as will be discussed in the methodology section, this makes the damping tail an excellent consistency check when fitting models. This high multipole region of the power spectra is where we start to see competing models diverge making it an extremely important area to measure.

Future observatories to probe these high multipoles are currently in the works. For example, CMB-S4 (cosmic microwave background stage four) is the next generation of CMB observations. It will consist of 12 ground-based aperture telescopes based in Chile and the south pole. Such telescopes have the aims to reach higher multipoles, revealing more of the damping tail.

Methodology

Exploring the effects of each parameter in Λ CDM on the power spectra:

Choosing different values for each cosmological parameter will produce a different curve for the power spectra. Comparing a curve produced by parameters predicted by a cosmological model to the data from the CMB will give a measure of how good that model is.

To see how the six parameters of Λ CDM affects the power spectra, code for anisotropies in the microwave background, CAMB, [16] was used to vary each parameter and plot the resulting power spectra. Code from the example notebook provided in the documentation was modified by looping over different values of each parameter to produce these plots.

First varying Ω_b .

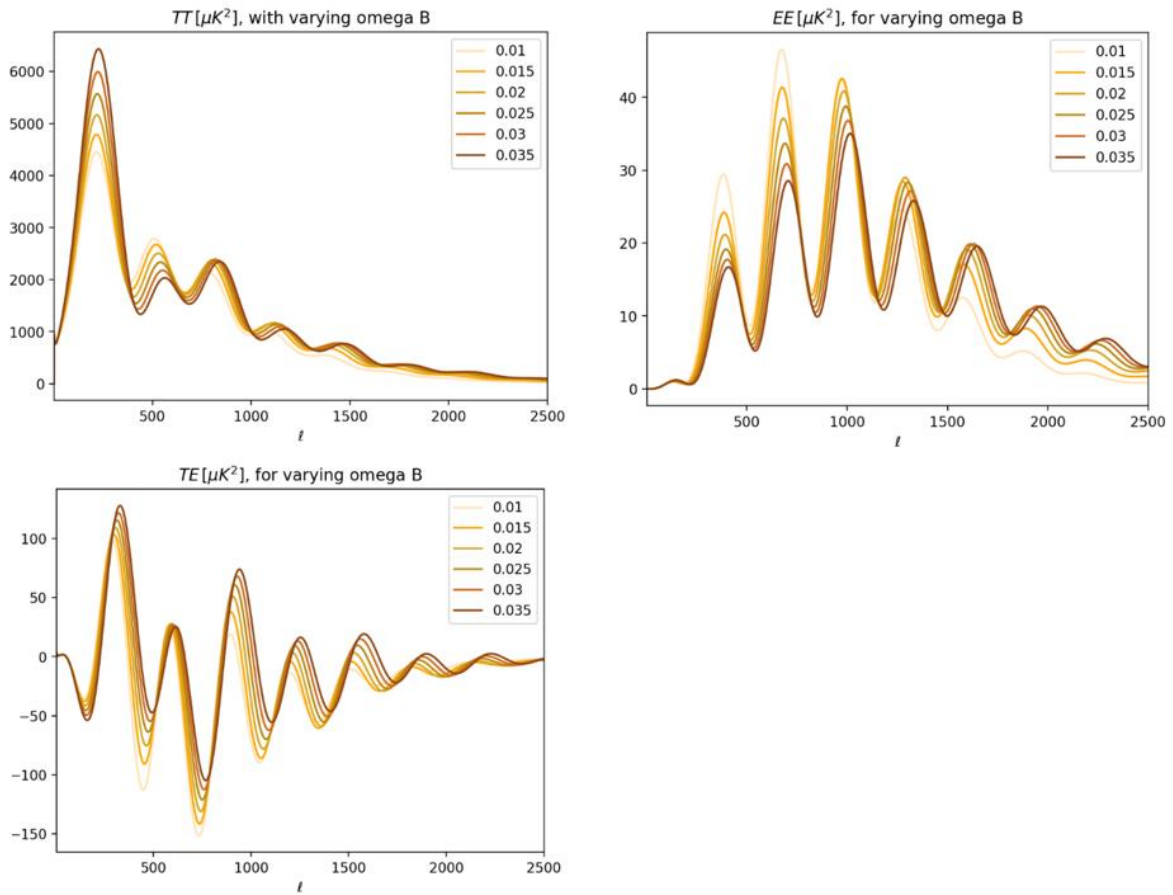


Figure 9: plots for the TT, EE and TE power spectra with varying values of Ω_b . Plotted using CAMB code.

From figure 9, it can be seen that, increasing the number of Baryons increases odd number peaks of the temperature-temperature (TT) spectra. This is because increasing the density of baryons deepens the potential wells by increasing gravitational force. This strengthens the in-fall part of the oscillation and reduces the radiation driven expansion section. Larger baryon density will increase the oscillation period (or decrease the frequency) and decrease damping. This can be explained using the mass on spring analogy. The height difference between the first and second peak represents the baryon density.

These plots have been matched with similar plots in literature [17]

Varying Ω_c .

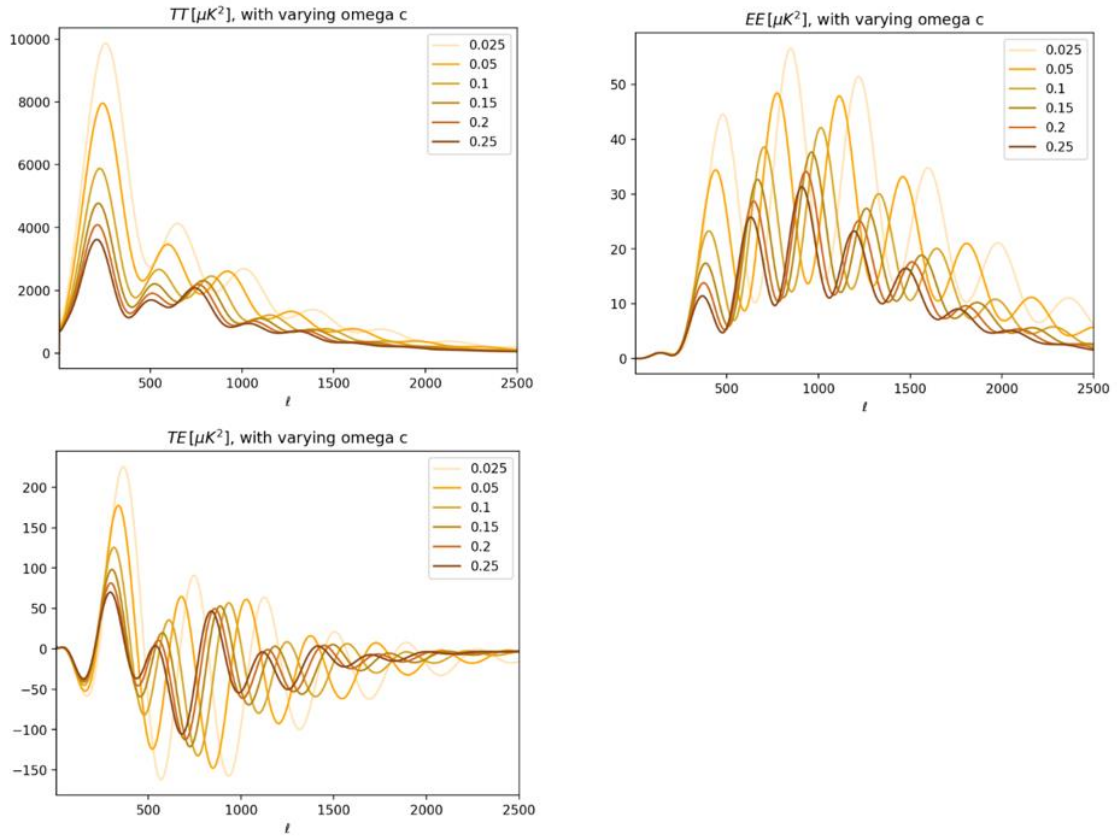


Figure 10: plots for the TT, EE and TE power spectra with varying values of Ω_c . Plotted using CAMB code.

From figure 10 it can be seen that increasing dark matter density decreases the amplitude of the peaks. Dark matter affects the time spent in the radiation dominated era. During this time, the main potential keeping the photon-baryon fluid oscillating is pressure. This is known as the driving effect. In the matter dominated era, pressure is no longer generating potential, so the driving effect no longer takes place. Increasing dark matter decreases the time spent in the radiation dominated era so decreases the amplitude of the peaks as the times spent with the driving effect in action decreases. This affects the higher multipole peaks the most as the smaller wavelengths complete the most oscillations in the radiation dominated era. The first and second peaks are mostly affected by gravitational affects so the variation in dark matter density can be mostly seen in the third peak. A third peak higher than the second indicates dark matter domination.

Varying n_s .

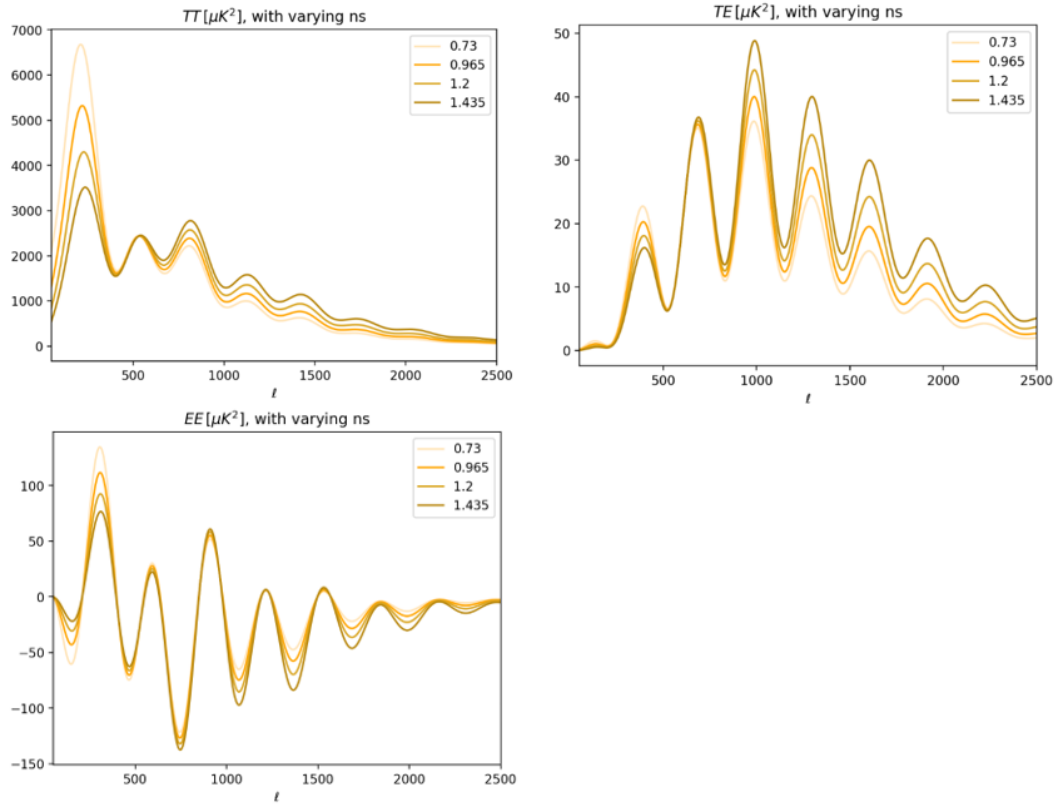


figure 11: plots for the TT, EE and TE power spectra with varying values of n_s . Plotted using CAMB code.

Scalar spectral index describes how density fluctuations vary with angular scale. Because density fluctuations were caused by quantum fluctuations which were intern caused during a period of rapid inflation the value of n_s tells us about how primordial fluctuations were distributed. From figure 11 we can see that varying n_s has the largest effect on the first peak of the power spectra. Specifically smaller values of n_s increasing its amplitude of the first peak as well as introducing a tilt.

Varying τ .

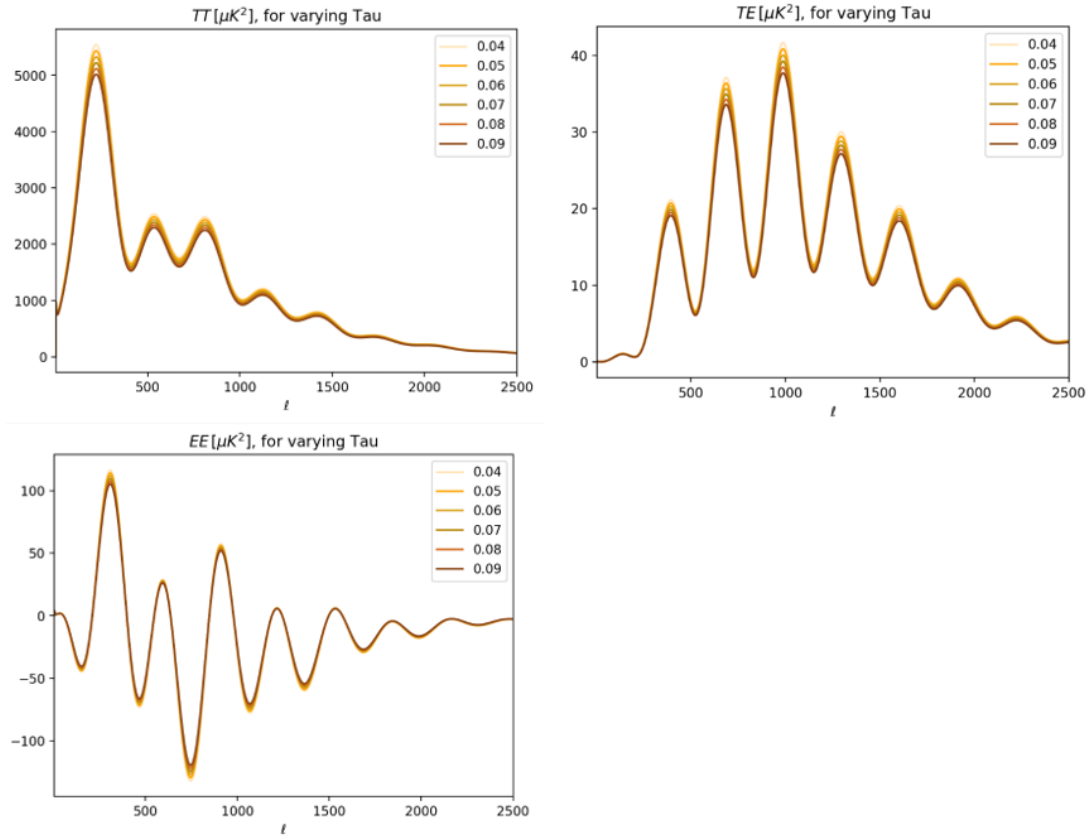


Figure 12: plots for the TT, EE and TE power spectra with varying values of τ . Plotted using CAMB code.

Reionisation is the period after CMB photons were released where galaxies first begin to form containing large stars which ionise the previously atomic hydrogen. These galaxies grew and ionised large pockets of hydrogen gas eventually ionising all the hydrogen in the universe. This is the point of reionisation. The CMB photons must travel through this ionised gas to reach Earth. As they do the photons scatter bending the path of the photons. This reduces the amplitude of the resulting signal we receive. As seen in figure 12, Increasing τ shifts the TT powers spectra downwards. Larger τ implies a that reionisation happened at a larger redshift or longer ago. This means that larger τ gives stars and galaxies a longer time to evolve.

Varying h.

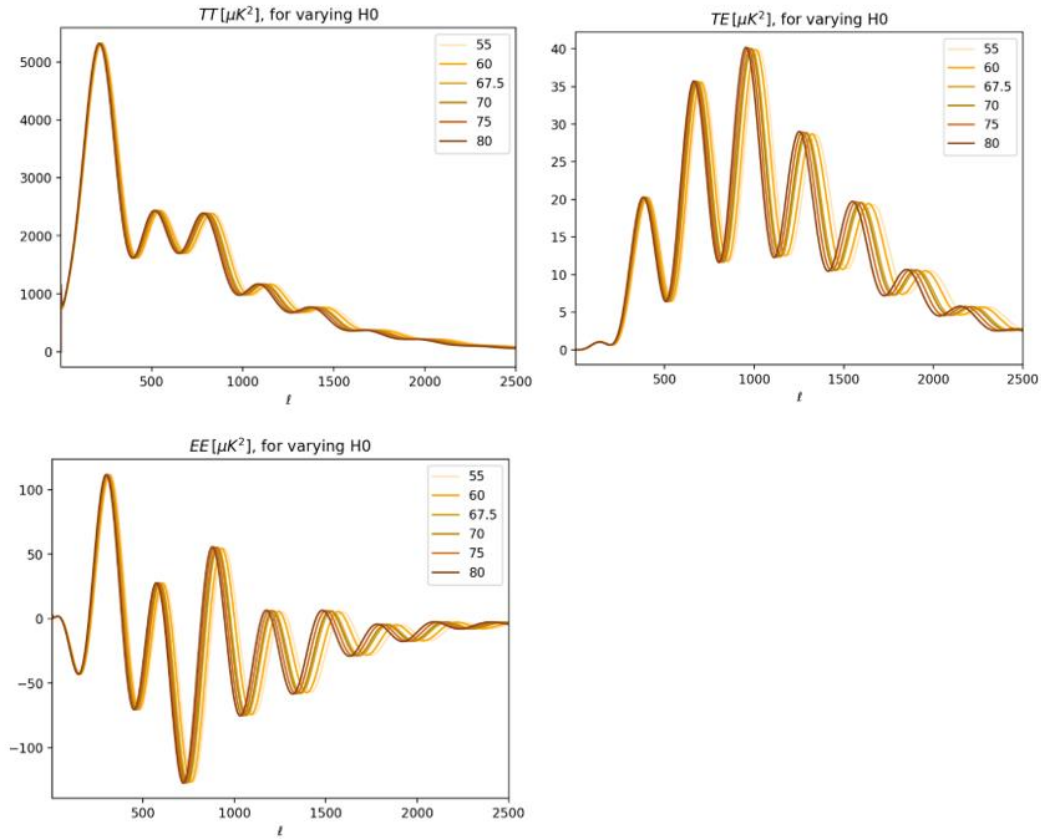


Figure 13: plots for the TT, EE and TE power spectra with varying values of H_0 . Plotted using CAMB code.

The Hubbel constant is a measure of how fast the universe is expanding. H_0 and h are directly related by $h = \frac{H_0}{100 \text{ km s}^{-1} \text{ Mpc}^{-1}}$. A larger value for H_0 means the universe is expanding faster. Figure 13 shows that a higher value for H_0 , or a faster expanding universe, shifts the TT power spectra to the left. Larger values for H_0 mean that the universe is expanding faster so at recombination, the universe would be larger.

Varying A_s .

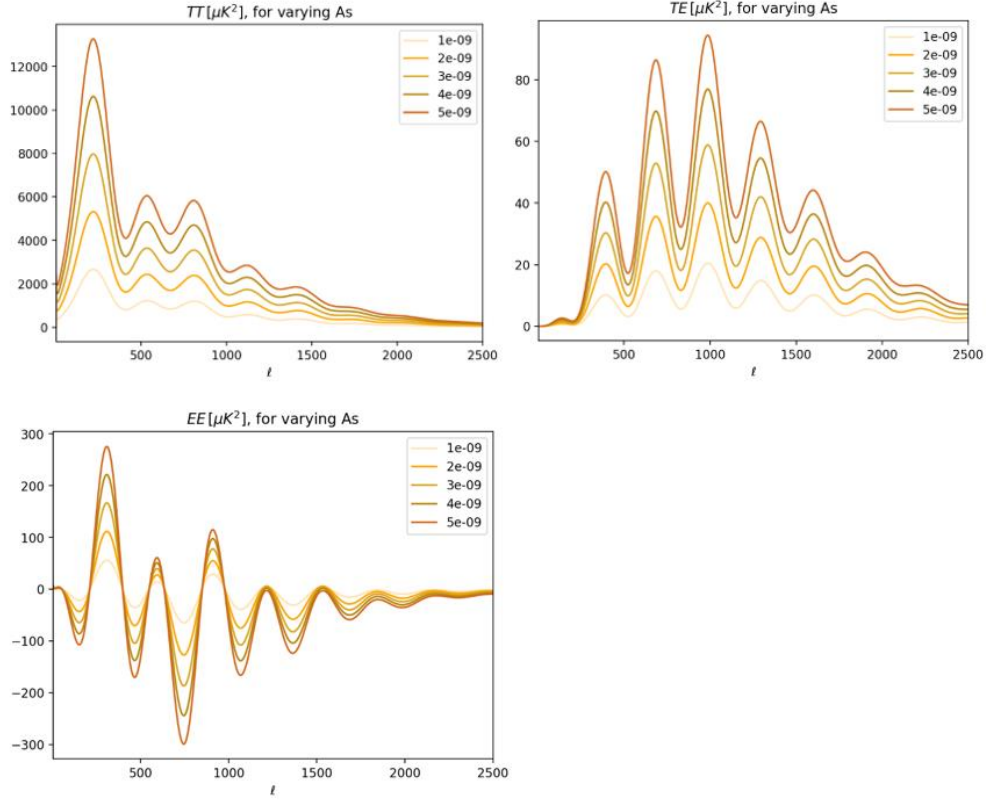


Figure 14: plots for the TT, EE and TE power spectra with varying values of A_s . Plotted using CAMB code.

A_s is the amplitude of fluctuations so by increasing it you will shift the TT power spectra upwards. This can be seen in figure 14.

This section also demonstrates how the code module CAMB can be used for fitting power spectra data. Because this code can produce a model based on any set of parameters its not hard to see that it could be used in reverse to determine a best fit set of parameters based on a data set.

What future observation of the CMB will reveal:

Now that the effects of the 6 key parameters in Λ CDM on the power spectra have been demonstrated and a useful code, CAMB, for CMB analysis has been introduced, the rest of this report will focus on the analysis of a hypothetical, best case data set that the next generation of CMB observatories will get very close to. This has been done with the aims of investigating what improvements could be made compared to Planck observations and to develop a MCMC algorithm which will be used for this analysis.

Planck only had the precision to reach multipoles of 2500 as discussed in the introduction section. This means only a small section of the dampening tail has been measured.

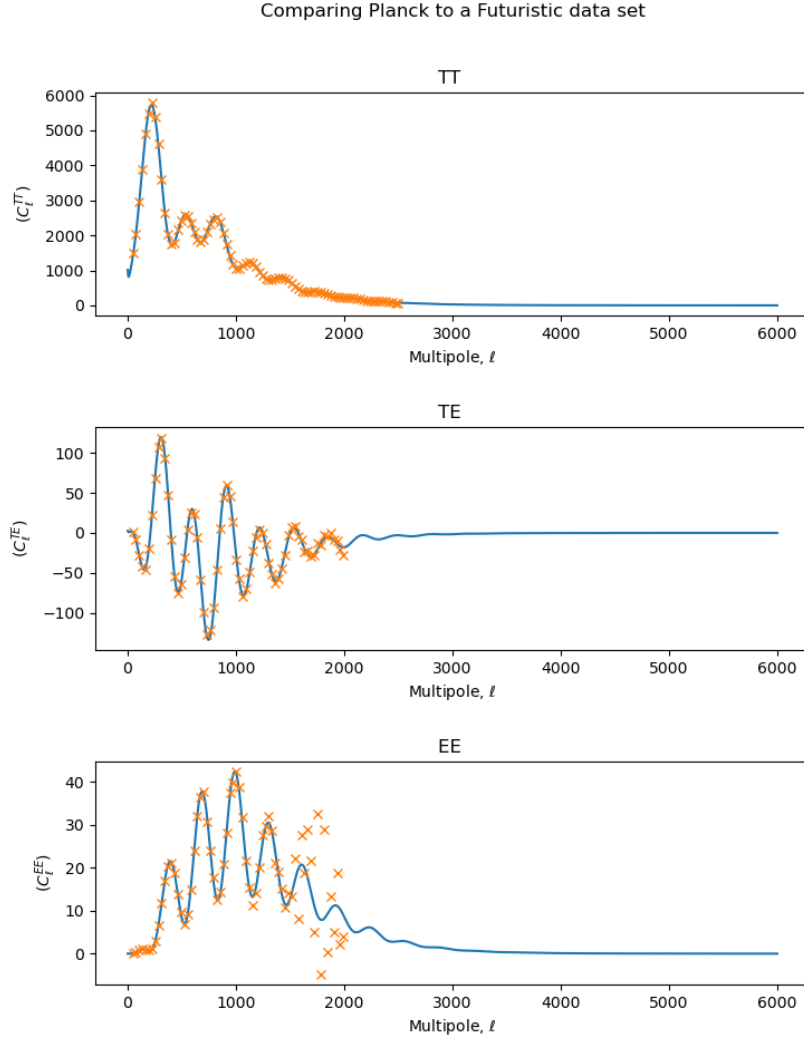


Figure 15: data points from the Planck power spectra TT (top), TE (middle) and EE (bottom) plotted as orange crosses over what could be achieved with future observations plotted as non-binned data in a blue line.

With future observations much higher precision could be achieved enabling much higher multipoles to be reached. Figure 15 shows how these observations would reveal much more of the dampening tail. An important part of the TT power spectrum for consistency checks. This is also where we start to see competing models of cosmology diverging from each other, making it an important region to explore. It can also be seen how future data would improve on measurements already made by Planck especially for the EE power spectra.

Observations of higher multipoles have already begun. Multiple ground-based telescopes such as the 10-meter south pole telescope (SPT) which has recently received a polarisation sensitive receiver with the aims of detecting B-mode polarisation. Other telescopes such as the Atacama cosmology telescope (ACT) and POLARBEAR are also observing the CMB.

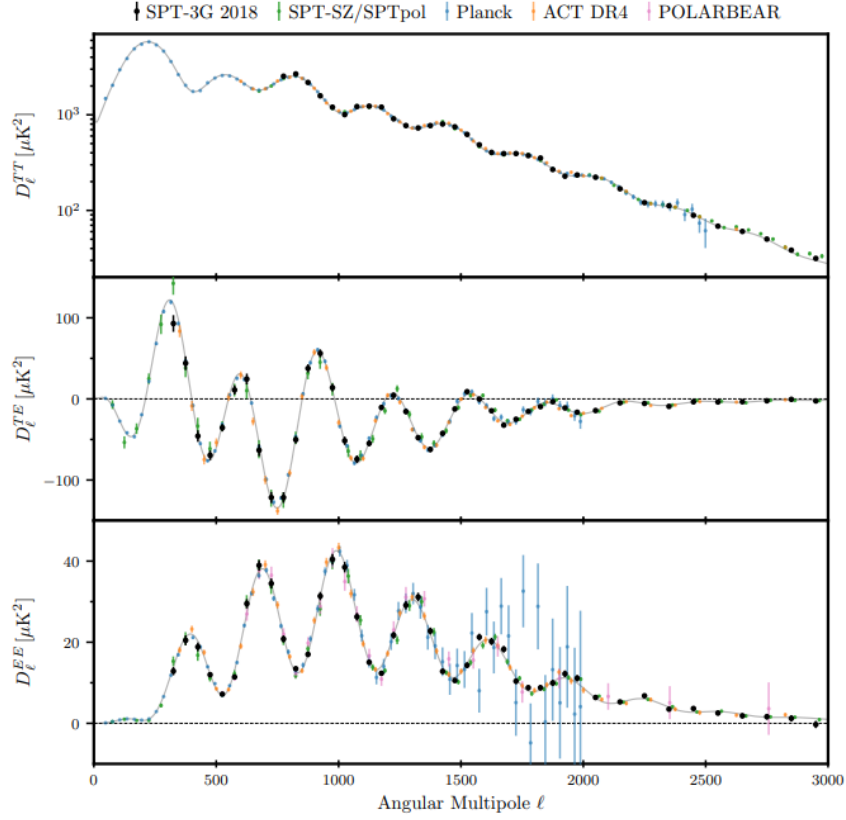


Figure 16: Data from ground based observatories SPT-3G, SPT-SZ, ACT and POLARBEAR compared to data from the Planck satellite for the TT (top), TE (middle) and EE (bottom) power spectra. L. Balkenhol et al. [18].

These experiments are already improving on the data from Planck. Figure 16 shows how data from these observations is reaching multipoles of 3000. It can also be seen how this data better constrains the power spectra for multipoles covered by Planck especially for the EE power spectra between multipoles of around 1500 to 2000.

Statistical analysis of the CMB:

We have discussed how important the power spectra are in testing models of cosmology. This section will discuss some of the statistical tools used to get a more concrete measure of how well a data set fits a model. Though residual plots are a good visual method of determining the fit of a model, the likelihood, L , gives a statistical measure of this. The likelihood function, often just called likelihood, is a probability density defined by a parameter which tells you how likely an outcome is to be observed assuming the defining parameter remains fixed. It can be used to tell you what values of the parameters which characterize a model are most likely to fit a given data set.

Once you obtain measurements for the power spectra of the CMB, you then want to fit a model of cosmology, likely Λ CDM as it is currently the best, to it. Using the likelihood will tell you which set of the six cosmological parameters of Λ CDM best characterise your measurements of the power spectra.

Often, when doing this kind of analysis, we work with the log likelihood as the mathematics becomes much simpler. Taking the log of the likelihood removes an exponential term and we can use the log likelihood in a similar way to the likelihood. When considering data with gaussian shaped errors the log

likelihood is related to the chi squared, χ^2 , value by a factor of negative two. The chi squared value of a point can be obtained by a simple formula,

$$\chi^2 = \sum \frac{(O - E)^2}{\sigma^2}$$

Where O is the observed data point, E is the expected point taken from the model and σ is the error. The chi squared value is related to the log likelihood by a factor of minus two,

$$\chi^2 = -2\ln(L)$$

So, we divide chi squared by negative two to obtain the log likelihood for that point. To obtain the total log likelihood for the whole data set we sum the likelihoods calculated for each point. The closer to zero the log likelihood is, the less scatter there is in the data.

Now we have a method for determining how well a model determined by a given set of parameters fits our observed data. To make this tool more useful, we can vary the parameters and determine which set best describes the data. When a model is characterised by only one parameter, this is simpler to do but when using models such as Λ CDM which is defined by six parameters, some of which have degeneracies, this can be more complicated. When performing this type of analysis on the CMB power spectra, researchers use a type of computer algorithm known as Markov chain Monte Carlo or MCMC.

Markov chain Monte Carlo.

Markov chain Monte Carlo algorithms approximate a probability distribution through sampling. As the name suggests, they are made up of two methods, Markov chains and Monte Carlo sampling. We will first briefly discuss the two constituent methods before tackling MCMC chains as a whole.

Monte Carlo sampling is a method of sampling from a probability distribution using randomness. These samples are then used to approximate a quantity. It is often used to approximate complex systems. A simple example is finding the area under a curve within a given quadrant. To do this you would uniformly distribute points throughout the quadrant and determine which are under the curve and which above. Using the ratio of total points below the curve to total points above the curve, you can approximate the area under the curve.

Markov chains describe a system of events where each event depends only on the event before it. Or in other words, predictions are made based only on the present state of the system.

MCMC chains combine these two methods with the aim to arrive at the target distribution and step around it to sample its distribution. A well implemented MCMC chain will spend an amount of time in each interval of the target distribution that is proportional to its density thus making a good approximation of it.

Though there are different ways to implement MCMC we will focus on a method which falls under the family of Metropolis-Hasting's algorithms and applying it to CMB power spectra analysis as this is most relevant to this report. The algorithm will begin with a starting set of parameters. These will be the six parameters from Λ CDM as we aim to fit our data of the CMB power spectra to a model based off Λ CDM. It will then generate the Λ CDM model and compare it to the observed data by calculating the log likelihood. Next it will take a random step in parameter space, it varies each parameter randomly, and calculates a new model with these new parameters, compares it to the observed data and calculates a log likelihood. The next step is quite important. The algorithm will compare the log likelihood of the initial model to the new model. If the new model has a log likelihood closer to zero, it will override the initial parameters with the new as this set has produced a better fit. If the new model produces a

likelihood further from zero than the initial model, it will keep the old set of parameters as these produced the better fit. The algorithm will then iterate through this cycle.

The first few iterations of the algorithm are known as burn in samples as they do not typically draw from the target distribution. Often these burn in samples are not considered in further analysis, but they are necessary in reaching the desired target distribution. Once the algorithm has reached the target distribution, it continues to sample it to produce the best approximation possible.

One of the key traits of MCMC chains is how, once it arrives at the target distribution, it does not leave it. This is why the Markov chain method is used. Because each point is only based on the last, once the algorithm reaches the target distribution, each subsequent sample is based on the target distribution.

MCMC analysis was designed to be used on probability distributions which are highly dimensional and therefore too complex to be solved analytically. This makes it suited to CMB analysis as we are comparing data to models defined by multiple parameters.

To further analyse the futuristic data set a Markov chain Monte Carlo algorithm was developed. The CAMB module was used to produce models from Λ CDM for different parameters and their goodness of fit was assessed using the log likelihood. Parameters from the Planck 2018 data release were used as a beginning point and bounds were imposed upon the random stepping vector proportional to the errors of those parameters.

To verify that the algorithm functioned properly the log likelihood for each step was plotted.

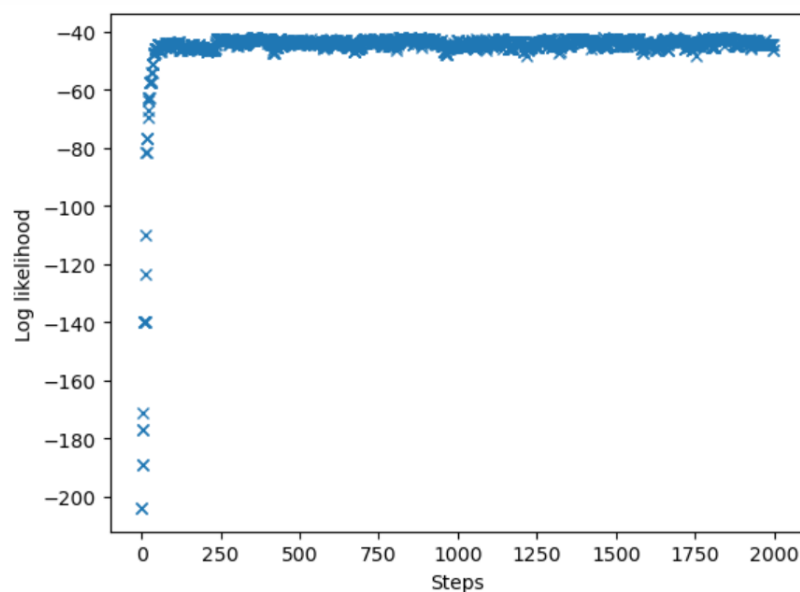


Figure 17: A plot of log likelihood vs steps for the MCMC algorithm fitting a Λ CDM model to the futuristic best case data set.

It is expected to see a number of burn in points with larger magnitude log likelihoods that approach zero as the algorithm makes steps. This can be seen in figure 17. The algorithm begins with a large negative value for log likelihood then, as it makes steps through parameter space, this value gets closer to zero. Once the log likelihood begins to stabilise the algorithm has reached the target distribution and begins to sample this. Figure 17 shows how the algorithm stabilizes just below a log likelihood of -40. This shows how the algorithm functions as it should and can be used for further analysis.

Contour plots.

Once you've performed an MCMC analysis a useful way of presenting and interpreting your analysis is through contour plots.

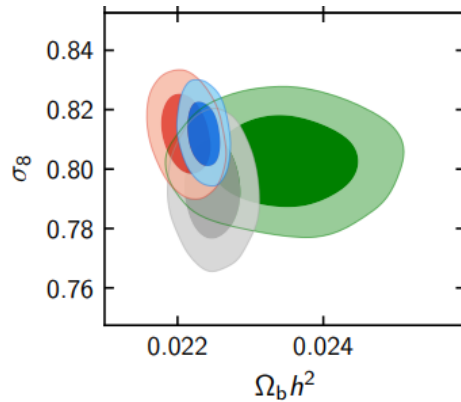


Figure 18: A contour plot showing how the density of the parameters σ_8 and Ω_b . The colours represent analysis of each power spectrum. Green is EE, grey TE, red TT, and blue is a combination of EE, TE and TT. The darkest of each colour contains 68% of the probability and the lightest contains 95%. Taken from the Planck 2018 results [13]

Figure 18 shows the contour plot for the parameters σ_8 and Ω_b . The darker the colour, the longer the MCMC chain spent in that region. As discussed previously, the longer the MCMC chain spends in a region the higher the probability density is. By plotting the analysis of each spectrum on the same graph but in different colours, it is easy to see how each overlap. This means the highest probability density is contained where each power spectra analysis overlaps. This is why the blue contour is smallest. It represents the combination of the analysis of the three spectra. This demonstrates how combining data sets can narrow down the probability density giving better constraints of parameters. This isn't only done with the CMB power spectra but also with CMB analysis and type Ia super nova for example.

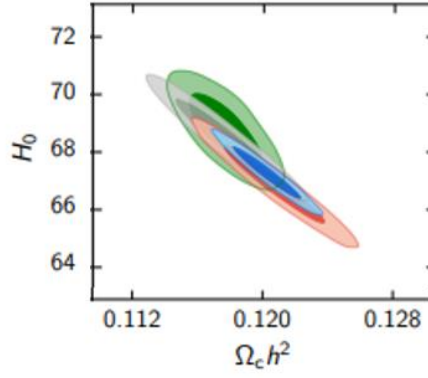


Figure 19: A contour plot for parameters H_0 and Ω_c . The colours represent analysis of each power spectrum. Green is EE, grey TE, red TT, and blue is a combination of EE, TE and TT. The darkest of each colour contains 68% of the probability and the lightest contains 95%. Taken from the Planck 2018 results [13]

Contour plots can also show degeneracies between particles. Figure 19 highlights that H_0 and Ω_c follow a linear relationship.

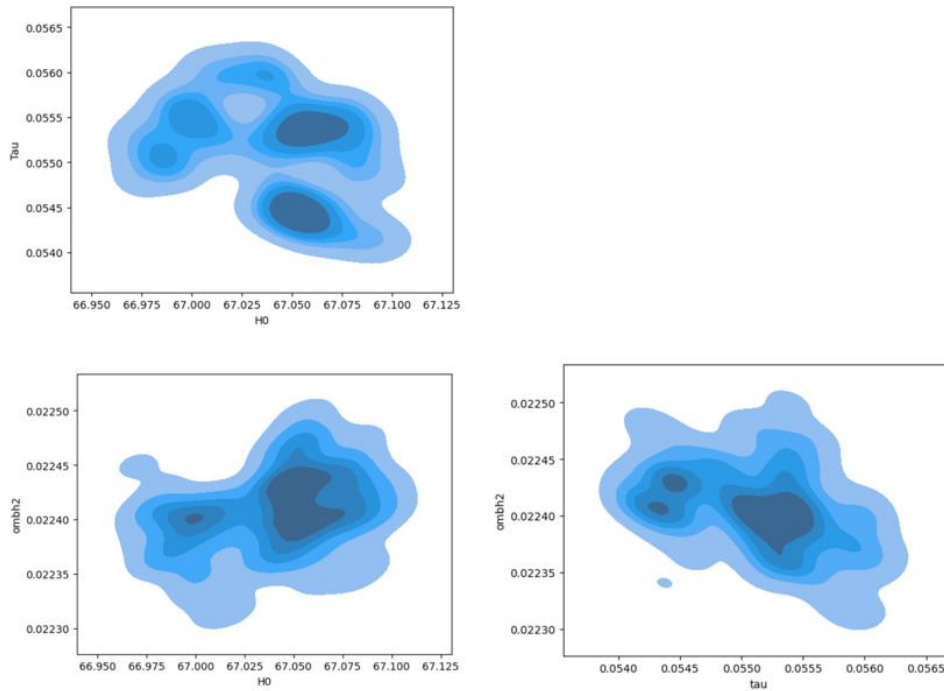


Figure 20: contour plots for parameters tau, τ , omb, Ω_b , and H_0 . Darker regions showing where the algorithm spent the most time.

With the chains from the MCMC algorithm, contour plots were plotted. Figure 20 shows the contour plots for τ , Ω_b and H_0 . To test their validity these chains were compared to chains from the Planck collaboration.

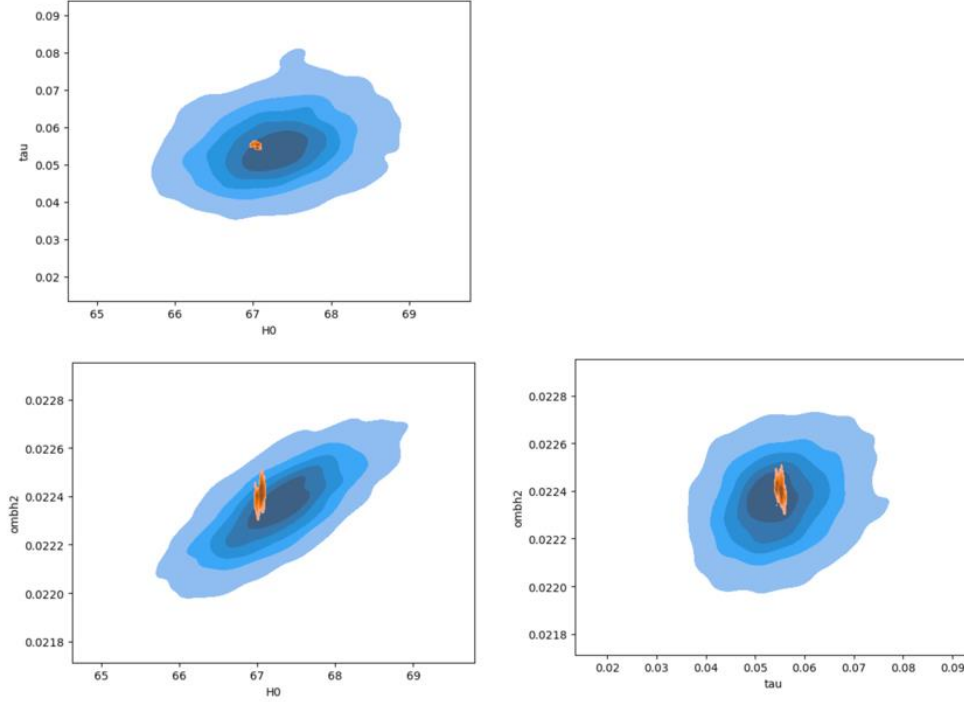


Figure 21: contour plots for parameters τ , Ω_b , and H_0 . Blue contours show data from Planck chains and Orange from the MCMC algorithm.

Figure 21 shows how each parameter falls within the expected ranges from the Planck data. It is also clear to see how much smaller the distributions are for the analysis of the futuristic data set. This is because the increased precision of observation will lead to better constraints on each parameter.

Residual plots

From The chain produced by the algorithm, the set of parameters for the likelihood closest to zero was taken. These were used in the CAMB module to produce a Λ CDM model which best fits the futuristic data set. This model was compared to the model produced by the Planck 2018 parameters. This was done by considering the residual plots using the futuristic data set and these two models.

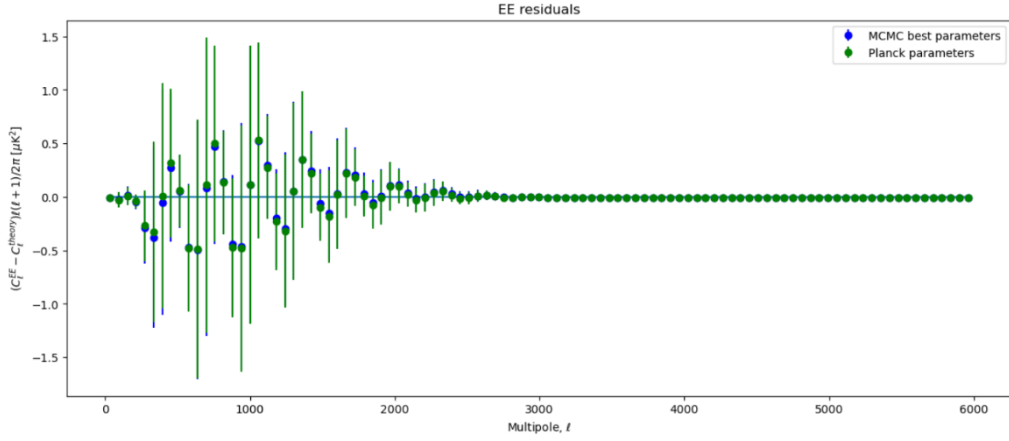


Figure 22: residual plots comparing the fits of the models produced using the Planck parameters, green, and the parameters form the MCMC best fit, blue for the EE power spectrum.

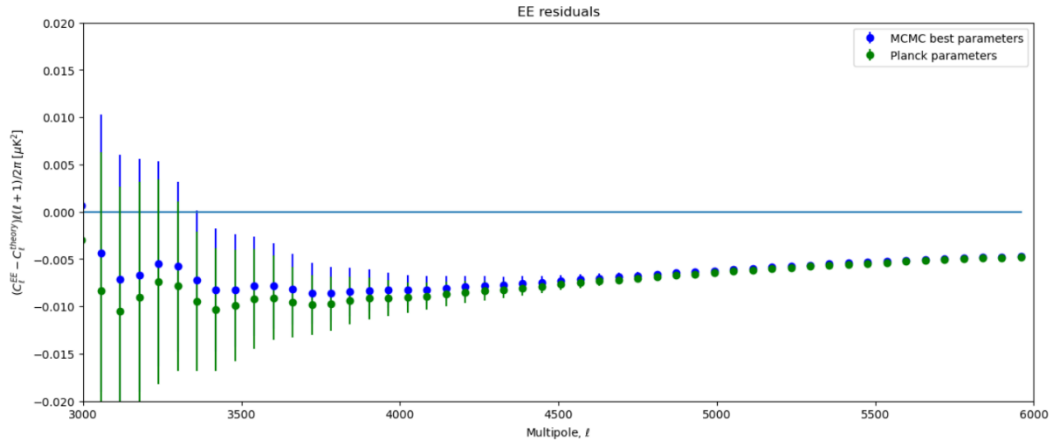


Figure 23: residual plots comparing the fits of the models produced using the Planck parameters, green, and the parameters form the MCMC best fit, blue for the EE power spectrum focusing on the high multipole range.

From figure 22 and 23 it can be seen how the futuristic data set has less scatter around its theory line. Though this is an improvement, these figures suggest that perhaps more tuning is needed.

Discussion:

By developing an understanding of what the power spectra of the CMB are and how they are used to constrain cosmological parameters, testing models of cosmology, analysis of a futuristic data set, which reflects what the next generation of telescope could achieve, has been carried out through development of a Markov chain Monte Carlo algorithm. This analysis was able to demonstrate how future observations could reach much higher multipoles than what Planck could and improve upon multipoles already measured by Planck. The MCMC algorithm was used to produce contour plots which were compared to that of Planck's and a best fit parameter set, which was used to demonstrate the improvement that could be made to the scatter of the power spectra, was compared to that of Planck.

References:

- [1] H. Bondi, T. Gold, 1948, The steady-state theory of the expanding universe
- [2] G. Gamow, 1948, The evolution of the universe, Nature vol 162
- [3] P.J.E. Peebles, 1965, The black-body radiation content of the universe and the formation of galaxies, The astrophysical journal volume 142 number 4.
- [4] R.H. Dike, P.J.E. Peebles, P.G. Roll and D.T. Wilkinson, 1965, Cosmic Black Body radiation,
- [5] Ruth Durrer, 2015, The cosmic microwave background: The history of its experimental investigation and its significance for cosmology, arxiv:1506.01907v1
- [6] P.A.R. Ade et al. 2014, Planck 2013 results. I. Overview of products and science results. Astronomy and Astrophysics.
- [7] Max Tegmark, accepted 1996, The angular power spectrum of the four year COBE data, The Astrophysical Journal 464:L35-L38
- [8] D. Larson et al., 2011, Seven-year Wilkinson microwave anisotropy probe (WMAP) observations: power spectra and WMAP-derived parameters, the astrophysical journal supplement series 192: 16 (19pp)
- [9] Marc Kamionkowski and Ely D. Kovetz, 2016, The Quest for B modes from Inflationary Gravitational waves, Annu. Rev. Astron. Astrophys. 54:227-69
- [10] J. Kovac et al., 2002, Detection of polarisation in the cosmic microwave background using DASI, Nature 420:772-787
- [11] P.A.R. Ade et al., 2017, Planck 2015 results. XX. Constrains on inflation, Astronomy and Astrophysics
- [12] Alan Heavens, 2008, The cosmological model: an overview and an outlook, Journal of Physics: conference series 120 022001
- [13] Planck Collaboration: N. Aghanim et. al., 2021, Planck 2018 results. VI. Cosmological parameters, Astronomy and Astrophysics.
- [14] Ronald J. Adler, Brendan Cassey and Ovid C. Jacob, 1995, Vacuum catastrophe: An elementary exposition of the cosmological constant problem.
- [15] White, S. D. M., Davis, M., & Frenk, C. S., 1984, The size of clusters in a neutrino-dominated universe, Monthly Notices of the Royal Astronomical Society (ISSN 0035-8711), vol. 209, July 15, 1984, p. 27P-31P. NASA-supported research.
- [16] Lewis Antony, Challinor Anthony, 2011, CAMB: code for anisotropies in the microwave background, Astrophysics Source Code Library, record ascl:1102.026
- [17] David Fernandez Bonet, 2020, Understanding the cosmic microwave background and its relevance in cosmology, university of Barcelona.
- [18] L. Balkenhol et. al., 2022, A measurement of the CMB temperature power spectrum and constraints on cosmology from the SPT-3G 2018 TT/TE/EE data set, arXiv:2212.05642.